

Electromagnetism - Question Bank

Course: Electromagnetism | B.Sc Physics | Semester 2 | Year 1

1. What is the SI unit of electric charge?

- a) Coulomb
- b) Volt
- c) Ampere
- d) Ohm

Answer: a

2. Coulomb's law states that the force between two charges is proportional to:

- a) the sum of the charges
- b) the product of the charges
- c) the difference of the charges
- d) the ratio of the charges

Answer: b

3. The SI unit of electric field is:

- a) newton metre
- b) volt per metre
- c) ampere per second
- d) watt

Answer: b

4. Outside a magnet, magnetic field lines run from:

- a) south pole to north pole
- b) north pole to south pole
- c) both directions
- d) nowhere

Answer: b

5. The force on a moving charge in a magnetic field is called the:

- a) Coulomb force
- b) Lorentz force
- c) frictional force
- d) tension force

Answer: b

6. A changing magnetic field through a coil induces:

- a) light
- b) a current
- c) heat
- d) sound

Answer: b

7. Faraday's law of induction is related to:

- a) resistance
- b) electromagnetic induction
- c) capacitance
- d) power factor

Answer: b

8. What is the SI unit of magnetic flux?

- a) Weber
- b) Tesla
- c) Henry
- d) Gauss

Answer: a

9. Ampere's law relates electric current to:

- a) electric field
- b) magnetic field
- c) electric potential
- d) charge density

Answer: b

10. The strength of an electromagnet increases with:

- a) fewer turns of wire
- b) more turns and more current
- c) shorter wire
- d) less current

Answer: b

11. State Faraday's law of electromagnetic induction.

(Write your answer in the space provided.)

12. Explain how a transformer works.

(Write your answer in the space provided.)